Minimal Point-Valued Stochastic Dynamics with Relaxing Memory

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We define a minimal point-valued stochastic dynamics on an abstract continuous state space Ω in which a visible trajectory x_n is coupled to an explicit exponentially relaxing memory density ρ_n. The projected process x_n is generally non-Markovian, whereas the augmented state $z_n = (x_n, \rho_n)$ is Markov by construction. The memory update compresses ordered histories into a current density and thereby defines an intrinsic arrow of update order without assuming spacetime, physical time, or thermodynamic irreversibility. Coarse-graining on the memory-persistence scale gives an effective parameter $t = \alpha n$ and, when a separation of scales exists, an approximate SDE/Fokker–Planck description. Within this effective regime we define metastable dynamical knots and coordinate-robust relaxation diagnostics. Any calibration to physical units, relativistic dynamics, or particle-physics structure is left to later work.

I. INTRODUCTION

Many approaches to fundamental physics begin with structures such as spacetime, locality, fields, or quantum states [1–5]. Here we ask a more limited question: what is obtained if one assumes only a sequence of non-trivial update events, a visible state, and an internal memory that relaxes but is not externally prescribed?

The construction is close in spirit to random processes with reinforcement and self-interacting walks or diffusions biased by their visitation history, including the true self-avoiding walk, self-interacting diffusions, and self-repelling Brownian polymers [6–12]. Its distinctive feature is that the retained history is not a full occupation measure or local time, but an explicit exponentially relaxing density, in the spirit of reduced memory variables in nonequilibrium dynamics [13, 14]. This gives a finite Markov embedding while preserving non-Markovian visible dynamics, consistent with standard finite-memory embeddings and with broader Markovian-embedding methods for memory equations [15–18].

The purpose of this paper is correspondingly narrow. We define the model, state the Markov embedding, extract the intrinsic coarse-grained time scale, and introduce metastability diagnostics for localized structures generated by the memory coupling. The paper does not assume a physical spacetime metric, conserved quantities, quantum postulates, or calibrated mass units.

II. MODEL AND MARKOV EMBEDDING

Let Ω be a continuous state space with a reference measure dx and local coordinates sufficient to write integrals and derivatives. The background coordinates are a mathematical scaffold for the generator and diagnostics, not a postulated physical spacetime. At update step n the full state is

$$z_n = (x_n, \rho_n), \tag{1}$$

where $x_n \in \Omega$ is the visible state and ρ_n is a non-negative memory density on Ω. The general finite-measure formulation uses an independent forgetting rate λ_m and deposition strength β. The normalized convention used below sets $\lambda_\mathrm{m} = \beta = \alpha$, fixes the overall memory scale, and makes $M_n(\mathcal{K})$ a visitation weight.

The visible update is

$$x_{n+1} = x_n + \varepsilon\xi_n - \eta\nabla\Phi_n(x_n), \tag{2}$$

where ξ_n is an independent isotropic unit-variance noise variable [19]. The memory evolves as

$$\rho_{n+1}(x) = (1 - \lambda_\mathrm{m})\rho_n(x) + \beta G_\sigma(x - x_{n+1}), \qquad 0 < \lambda_\mathrm{m} < 1. \tag{3}$$

with $\beta \geq 0$ and G_σ a smooth normalized kernel of width σ. In the normalized convention $\lambda_\mathrm{m} = \beta = \alpha$, this reduces to the update used in the figures and scaling discussion. The memory-induced potential is

$$\Phi_n(x) = \int_\Omega K(x, y)\rho_n(y)\, dy. \tag{4}$$

The gradient in Eq. (2) acts on the first argument of K:

$$\nabla\Phi_n(x) = \int_\Omega \nabla_x K(x, y)\rho_n(y)\, dy, \tag{5}$$

whenever differentiating under the integral is justified. For a translation-invariant kernel $K(x, y) = K_\ell(x - y)$ this is the convolution of ρ_n with ∇K_ℓ.

Given z_n and a fresh draw ξ_n, Eqs. (2)–(3) determine the distribution of z_{n+1}. Thus the sequence $\{z_n\}_{n\geq 0}$ is Markov. The visible sequence $\{x_n\}_{n\geq 0}$ alone is generally non-Markovian, because identical visible states can have different retained memory fields and hence different next-step laws. This is the basic Markov embedding of the model.

To make this embedding explicit, define the transition kernel $P(z, A) = \mathbb{P}\{z_{n+1} \in A \mid z_n = z\}$ on the augmented state space Σ. On bounded observables this kernel acts by the positive unital Markov/Koopman operator

$$(Uf)(z) = \int_\Sigma f(z')P(z, dz') = \mathbb{E}[f(z_{n+1}) \mid z_n = z]. \tag{6}$$

The iterates form a semigroup, $U^{m+n} = U^m U^n$. Because of noise and history compression this is not, in general, a reversible group action or a deterministic algebra automorphism; in stochastic settings $U(fg)$ need not equal $(Uf)(Ug)$. This places the model in the same operator-theoretic language as stochastic Koopman/transfer-operator methods, while related algebraic formulations view stochastic maps as positive unital maps on commutative C^*-algebras or as morphisms in Markov categories [20–23].

In the normalized convention, Eq. (3) also explains the arrow of update order. It maps an ordered trajectory history to a current density by exponentially suppressing older contributions. For fixed x_{n+1} the affine update can be inverted as a map in ρ_n, but the inverse does not recover the discarded ordered sequence of past positions. The irreversibility used here is therefore history compression, not thermodynamic entropy production.

Iterating the normalized form of Eq. (3) gives, for $n > k$,

$$\rho_n(x) = (1-\alpha)^{n-k}\rho_k(x) + \alpha \sum_{j=k+1}^{n} (1-\alpha)^{n-j} G_\sigma(x - x_j). \tag{7}$$

Thus a contribution of age $m = n - j$ enters with weight $\alpha(1-\alpha)^m \simeq \alpha e^{-\alpha m}$ for $\alpha \ll 1$. The persistence scale is of order α^{-1} update steps.

III. EFFECTIVE TIME

The update index n is not physical time. Nevertheless, the memory relaxation rate supplies an internal coarse-graining scale. When observables vary slowly on the persistence scale, it is natural to define the dimensionless parameter

$$t = \alpha n. \tag{8}$$

This parameter counts persistence-scale units along the update sequence. In a joint limit $\alpha \to 0$ and $\varepsilon \to 0$ at fixed ε^2/α, the discrete update may be summarized by the coarse-grained SDE

$$\mathrm{d}x = -\frac{\eta}{\alpha}\nabla\Phi[\rho(t)](x)\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}W_t, \qquad D \sim \frac{\varepsilon^2}{2\alpha}, \tag{9}$$

with convention-dependent numerical prefactors [24, 25]. The potential remains memory-induced and generally slowly time-dependent. In metastable regimes it can be treated as quasi-stationary on intermediate relaxation scales.

Operationally, α can be estimated from the exponential age weights in Eq. (7) or from the decay of a memory-sensitive observable such as $\Phi[\rho_n](x_n)$. The role of t is therefore not to postulate a physical clock, but to provide a dimensionless comparison parameter intrinsic to the relaxing-memory dynamics.

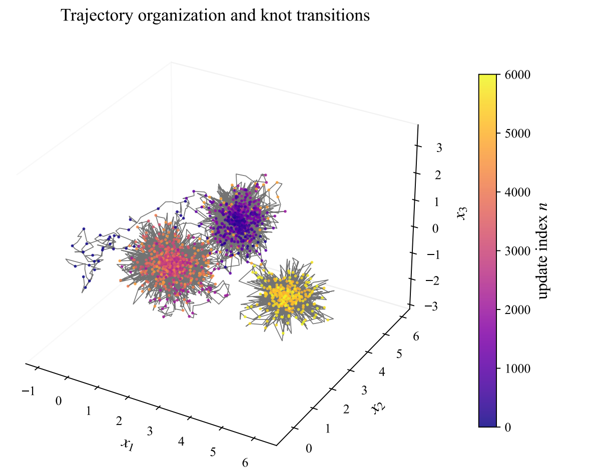

FIG. 1. Representative trajectory with multiple metastable knot-like regions. The line shows the visible trajectory, sampled points contribute to the memory density, and color denotes update index.

IV. DYNAMICAL KNOTS

The feedback between trajectory and memory can generate localized regions of repeated visitation. We call such metastable regions *dynamical knots*. They are not imposed fixed points or particles; they are persistent trajectory-memory patterns. Figure 1 shows a representative simulated trajectory in which distinct knot-like regions appear.

A knot can be identified operationally by a region $\mathcal{K} \subset \Omega$ whose memory/visitation weight

$$M_n(\mathcal{K}) = \int_{\mathcal{K}} \rho_n(x)\,dx \tag{10}$$

remains elevated over intervals $\Delta n \gg \alpha^{-1}$. Elevated $M_n(\mathcal{K})$ means that the memory density carries substantial weight near $\mathcal{K}$; through Eq. (4) this can generate a locally restoring drift that stabilizes the region against stochastic excursions. With augmented trajectory features, this criterion has a direct operator-theoretic version: metastable knots should appear as almost-invariant sets or slowly decaying modes of an estimated transfer operator at a chosen lag [15, 26]. The present paper uses residence and relaxation diagnostics as the minimal trajectory-level version of this test.

In a frozen-memory approximation, knots can be represented locally by points or regions where

$$\nabla\Phi[\rho](x_\star) = 0. \tag{11}$$

Linearizing the continuum drift near such a point yields an OU-type approximation

$$\mathrm{d}x = -\gamma H(x - x_\star)\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}W_t, \qquad \gamma = \frac{\eta}{\alpha}, \tag{12}$$

where H is the Hessian of the memory potential at $x_\star$. Local stability requires the symmetric part of H to be positive in the relevant directions [27]. If λ denotes a characteristic restoring eigenvalue, the relaxation rate scales as

$$\Gamma_{\mathrm{rel}} \sim \frac{\eta}{\alpha}\lambda. \tag{13}$$

V. RELAXATION DIAGNOSTICS

The stability of a dynamical knot is measured by how quickly perturbations relax back into the metastable region. Rare transitions are expected in the usual metastable regime of noisy dynamics [28, 29]. Let τ_{rel} be extracted, for example, from an in-knot autocorrelation $C(t) \propto e^{-t/\tau_{\mathrm{rel}}}$ or from the dominant relaxation eigenvalue of the linearized Fokker–Planck operator. We define

$$\Gamma_{\mathrm{rel}} = \frac{1}{\tau_{\mathrm{rel}}}. \tag{14}$$

Thus the diagnostic is a fitted decay scale of a coarse-grained trajectory observable, not a new microscopic parameter. In a transfer-operator description with lag τ, a slow eigenvalue $\mu_i(\tau)$ gives the analogous relaxation rate

$$\Gamma_i(\tau) = -\frac{1}{\tau}\log|\mu_i(\tau)|, \tag{15}$$

where $\mu_i(\tau)$ is a slow eigenvalue of the lagged operator. The local Hessian estimate in Eq. (13) is therefore a quasi-stationary local approximation to a broader spectral diagnostic.

Because absolute coordinate scales on Ω are not physically fixed, it is useful to report the dimensionless combination

$$\widehat{\Gamma} = \frac{\sigma^2}{D}\Gamma_{\mathrm{rel}}. \tag{16}$$

This compares relaxation to diffusion on the coarse-graining scale and is approximately invariant under uniform coordinate rescalings $x \mapsto ax$ with $\sigma \mapsto a\sigma$. A second useful coarse-grained scale is

$$Q_{\mathrm{rel}} = 2D\Gamma_{\mathrm{rel}}, \tag{17}$$

which summarizes stochastic broadening and relaxation without assigning physical energy units. A later calibration may determine whether a normalized version of this quantity can be interpreted as energy-like.

In this operational sense, a mass-like proxy is associated with relaxation or confinement strength rather than with a fundamental inertial parameter. Heavier structures correspond to faster return to the metastable region and/or steeper memory-induced restoring potentials. This terminology is only an effective-description shorthand; no identification with mc^2 or a relativistic mass scale is assumed here.

VI. DISCUSSION

We have introduced a minimal stochastic dynamics in which a visible point-valued state is coupled to an explicitly relaxing memory density. The visible dynamics is generally non-Markovian, while the augmented state (x_n, ρ_n) gives a natural Markov embedding. The same memory update compresses ordered histories and therefore supplies an intrinsic update direction. Coarse-graining on the memory-persistence scale yields the dimensionless parameter $t = \alpha n$ and, in suitable regimes, an approximate SDE/Fokker–Planck description.

Within this effective description, memory feedback can generate metastable dynamical knots. Their stability can be characterized by residence weights, relaxation rates, and dimensionless diagnostics such as $\widehat{\Gamma}$. These quantities are extractable from simulation trajectories and do not require physical spacetime units.

The construction remains intentionally limited. The continuum SDE is an effective approximation rather than a proven scaling theorem, and the existence and classification of knot regimes depend on the qualitative properties of (G_σ, K). Multiple coupled generators, emergent geometry, universal constants, quantum normal forms, and Standard-Model-like structures are natural next questions, but they are not assumptions or conclusions of the present minimal model. From the algebraic viewpoint, later work can ask which transformations preserve the transition kernel or its metastable sets; such stabilizer or equivariance structures are the appropriate place to look for effective symmetry algebras.

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[1] S. Weinberg. Ultraviolet divergences in quantum theories of gravitation. In S. W. Hawking and W. Israel, editors, *General Relativity: An Einstein Centenary Survey*, pages 790–831. Cambridge University Press, 1979.

[2] M. Reuter. Nonperturbative evolution equation for quantum gravity. *Phys. Rev. D*, 57:971–985, 1998.

[3] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin. Spacetime as a causal set. *Phys. Rev. Lett.*, 59:521–524, 1987.

[4] C. Rovelli. *Quantum Gravity.* Cambridge University Press, 2004.

[5] J. Polchinski. *String Theory. Volume I: An Introduction to the Bosonic String*. Cambridge University Press, 1998.

[6] R. Pemantle. A survey of random processes with reinforcement. *Probability Surveys*, 4:1–79, 2007.

[7] D. J. Amit, G. Parisi, and L. Peliti. Asymptotic behavior of the true self-avoiding walk. *Phys. Rev. B*, 27:1635–1645, 1983.

[8] B. Tóth and W. Werner. The true self-repelling motion. *Probab. Theory Relat. Fields*, 111:375–452, 1998.

[9] M. Benaïm and O. Raimond. Self-interacting diffusions. *Probability Theory and Related Fields*, 122:1–41, 2002.

[10] R. T. Durrett and L. C. G. Rogers. Asymptotic behavior of brownian polymers. *Probab. Theory Relat. Fields*, 92 (3):337–349, 1992.

[11] L. Peliti and L. Pietronero. Random walks with memory. *La Rivista del Nuovo Cimento*, 10(6):1–33, 1987.

[12] Simon Holbach and Olivier Raimond. Convergence to the uniform distribution of moderately self-interacting diffusions on compact Riemannian manifolds, 2026.

[13] H. Mori. Transport, collective motion, and brownian motion. *Prog. Theor. Phys.*, 33:423–455, 1965.

[14] R. Zwanzig. *Nonequilibrium Statistical Mechanics*. Oxford University Press, 2001.

[15] Frank Noé and Feliks Nüske. A variational approach to modeling slow processes in stochastic dynamical systems. *Multiscale Modeling & Simulation*, 11(2):635–655, 2013. doi:10.1137/110858616.

[16] Frank Noé, Hao Wu, Jan-Hendrik Prinz, and Nuria Plattner. Projected and hidden Markov models for calculating kinetics and metastable states of complex molecules. *The Journal of Chemical Physics*, 139(18):184114, 2013. doi:10.1063/1.4828816.

[17] Divya Jaganathan and Rahil N. Valani. Markovian embedding of nonlinear memory via spectral representation. *Communications in Nonlinear Science and Numerical Simulation*, 2025. doi:10.1016/j.cnsns.2025.109540.

[18] Mateusz Wiśniewski, Jerzy Łuczka, and Jakub Spiechowicz. Memory corrections to Markovian Langevin dynamics, 2024.

[19] C. Gardiner. *Stochastic Methods*. Springer, 2009.

[20] Matthew O. Williams, Ioannis G. Kevrekidis, and Clarence W. Rowley. A data-driven approximation of the Koopman operator: Extending dynamic mode decomposition. *Journal of Nonlinear Science*, 25:1307–1346, 2015. doi:10.1007/s00332-015-9258-5.

[21] Stefan Klus, Feliks Nüske, Péter Koltai, Hao Wu, Ioannis Kevrekidis, Christof Schütte, and Frank Noé. Data-driven model reduction and transfer operator approximation. *Journal of Nonlinear Science*, 28(3):985–1010, 2018. doi:10.1007/s00332-017-9437-7.

[22] Arthur J. Parzygnat. Discrete probabilistic and algebraic dynamics: a stochastic commutative Gelfand-Naimark theorem, 2017.

[23] Tobias Fritz. A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics. *Advances in Mathematics*, 370:107239, 2020. doi:10.1016/j.aim.2020.107239.

[24] B. Øksendal. *Stochastic Differential Equations: An Introduction with Applications*. Springer, 6 edition, 2003.

[25] H. Risken. *The Fokker–Planck Equation: Methods of Solution and Applications*. Springer, 2 edition, 1996.

[26] Gary Froyland, Simon Lloyd, and Naratip Santitissadeekorn. Coherent sets for nonautonomous dynamical systems. *Physica D: Nonlinear Phenomena*, 239(16): 1527–1541, 2010. doi:10.1016/j.physd.2010.03.009.

[27] S. Strogatz. *Nonlinear Dynamics and Chaos*. CRC Press, 2018.

[28] H. A. Kramers. Brownian motion in a field of force and the diffusion model of chemical reactions. *Physica*, 7(4): 284–304, 1940.

[29] M. Freidlin and A. Wentzell. *Random Perturbations of Dynamical Systems*. Springer, 3 edition, 2012.